

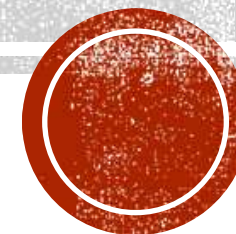
ECO-HEALTH WORKSHOP: CAUSAL MEDIATION ANALYSIS IN CLIMATE CHANGE EPIDEMIOLOGY PART II

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Tarik Benmarhnia

University of California, San Diego

UC Center for Climate Health and Equity



OUTLINE

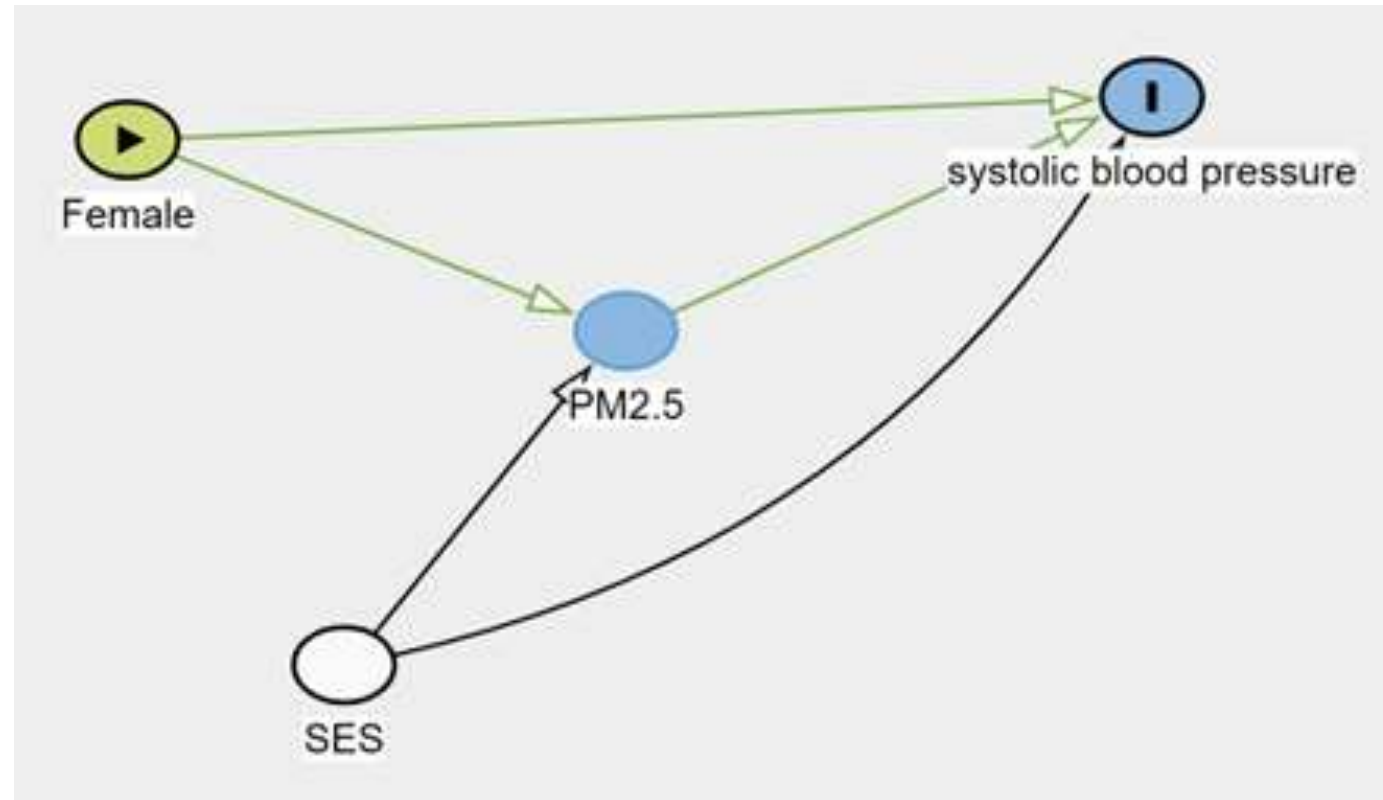
- Using mediation analyses for environmental justice applications
- Open Q&A



**APPLICATION:
GENDER INEQUALITIES IN BLOOD PRESSURE
AND THE ROLE OF FINE PARTICLES**

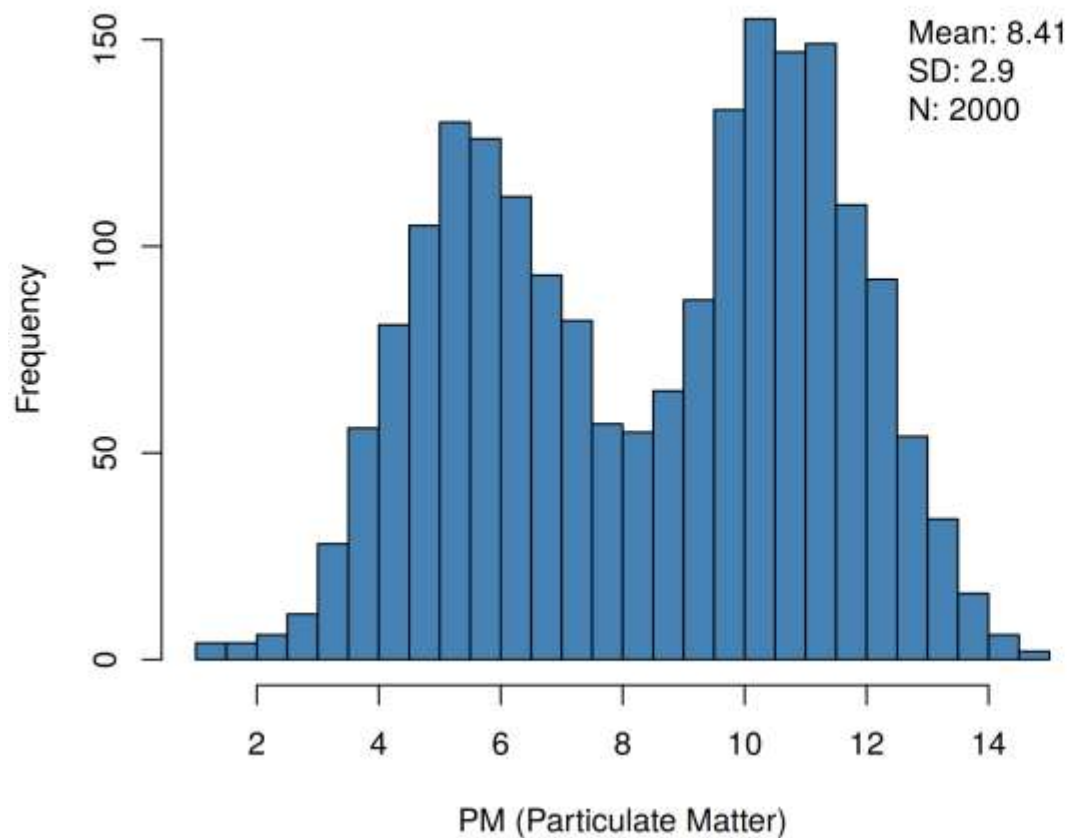


THE SETTING

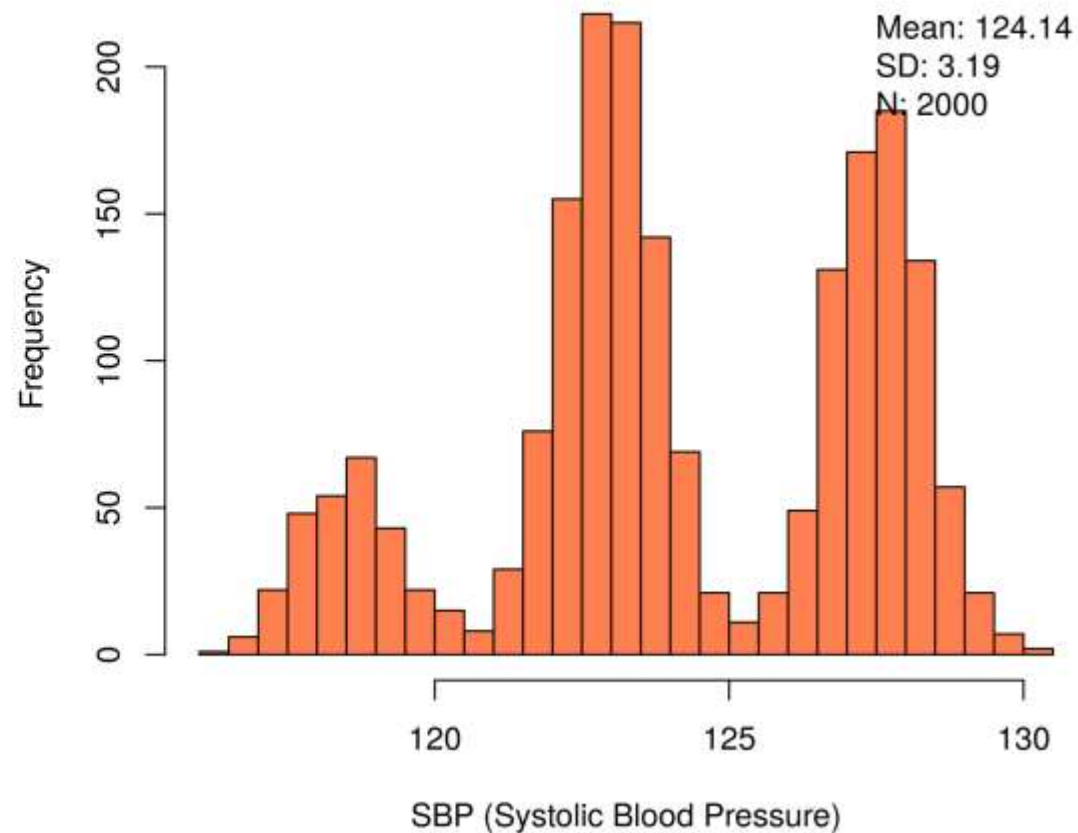


DESCRIPTIVE ANALYSES

Distribution of Particulate Matter Exposure



Distribution of Systolic Blood Pressure



Model Specifications

Outcome Model

$$SBP = \beta_0^y + \beta_1' \text{Female} + \beta_2' \text{PM} + \beta_3 \text{SES} + \varepsilon^y$$

Mediator Model

$$\text{PM} = \beta_0^M + \beta_1 \text{Female} + \varepsilon^M$$



Effect Decomposition

Natural Indirect Effect (NIE)

$$\text{NIE} = \beta_1 \times \beta_2'$$

Natural Direct Effect (NDE)

$$\text{NDE} = \beta_1'$$

Total Effect (TE)

$$\text{TE} = \text{NDE} + \text{NIE} = \beta_1' + (\beta_1 \times \beta_2')$$

Proportion Mediated

$$\text{PM} = \text{NIE} / \text{TE} = (\beta_1 \times \beta_2') / (\beta_1' + \beta_1 \times \beta_2')$$



ASSUMPTIONS IN MEDIATION ANALYSES

- We will consider 4 different assumptions that are compulsory to estimate causal effects.
 - From VanderWeele 2015; *Explanation in Causal Inference: Methods for Mediation and Interaction*
 - A1: Exposure – Outcome confounding
 - A2: Mediator – Outcome confounding
 - A3: Exposure – Mediator confounding
- Must be controlled and we assume
NO unmeasured confounders**
- A4: NO mediator – Outcome confounder itself affected by the exposure



THE PRODUCT METHOD

- Fit Models using GLM

- Outcome model: $SBP \sim Female + PM + SES$
`Sy.mod1 <- glm(sbp ~ female + pm + ses, data=dt, family="gaussian")`
- Mediator model: $PM \sim Female + SES$
`Sm.mod1 <- glm(pm ~ female + ses, data=dt, family="gaussian")`
- Total effect model: $SBP \sim Female + SES$ # (includes SES as pre-exposure confounder to estimate total effect)
`t.mod1 <- glm(sbp ~ female + ses, data=dt, family="gaussian")`

- Extract Point Estimates

- Direct Effect (NDE) `del <- coef(summary(y.mod1))["female", 1]`
- Indirect Effect (NIE) `iel <- coef(summary(y.mod1))["pm", 1] * coef(summary(m.mod1))["female", 1]`
- Total Effect (TE) `tel <- coef(summary(t.mod1))["female", 1]`
- Proportion Mediated (PM%) `pm1 <- iel / tel`
- TE from sum (verification) `te_sum <- del + iel`



GETTING CONFIDENCE INTERVALS USING BOOTSTRAPPING

```
#####  
# Bootstrap Function for Confidence Intervals  
#####  
  
mediation_boot_glm <- function(data, indices) {  
  # Resample data  
  boot_data <- data[indices, ]  
  
  # Fit models  
  m.mod <- glm(pm ~ female + ses, data=boot_data, family="gaussian")  
  y.mod <- glm(sbp ~ female + pm + ses, data=boot_data, family="gaussian")  
  t.mod <- glm(sbp ~ female + ses, data=boot_data, family="gaussian")  
  
  # Extract coefficients  
  de <- coef(summary(y.mod))["female", 1]  
  ie <- coef(summary(y.mod))["pm", 1] * coef(summary(m.mod))["female", 1]  
  te <- coef(summary(t.mod))["female", 1]  
  pm <- ie / te  
  
  return(c(NDE = de, NIE = ie, TE = te, PM = pm))  
}
```

```
#####  
# Run Bootstrap (1000 replicates)  
#####  
  
set.seed(123)  
cat("Running bootstrap (1000 replicates)...\n")  
boot_results <- boot(data = dt,  
                     statistic = mediation_boot_glm,  
                     R = 1000)  
  
# Extract 95% confidence intervals (percentile method)  
nde_ci <- boot.ci(boot_results, type = "perc", index = 1)  
nie_ci <- boot.ci(boot_results, type = "perc", index = 2)  
te_ci <- boot.ci(boot_results, type = "perc", index = 3)  
pm_ci <- boot.ci(boot_results, type = "perc", index = 4)  
#####
```

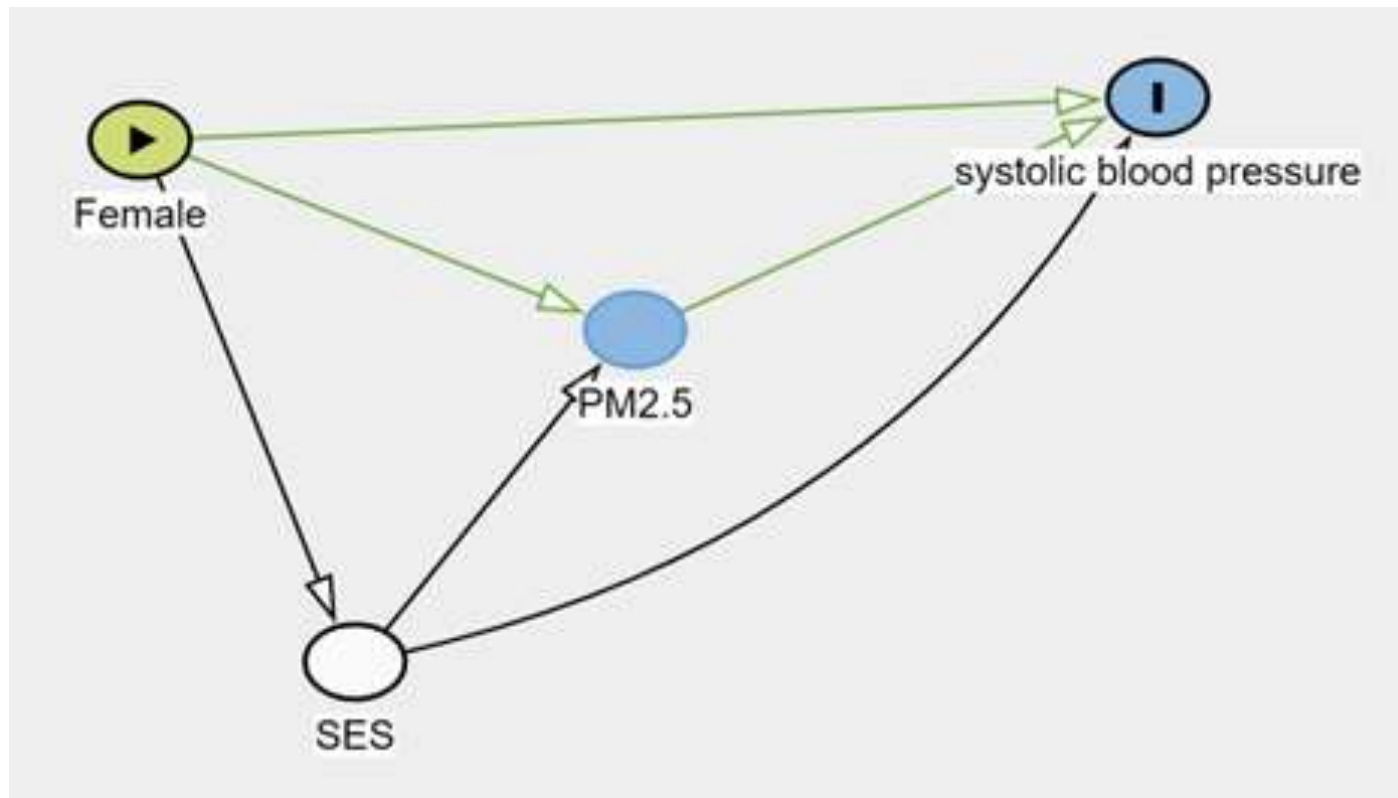


RESULTS

Effect	<i>Truth</i>	Estimate	95% CI
Natural Direct Effect (NDE)	2	2.007	(1.912, 2.098)
Natural Indirect Effect (NIE)	2.5	2.53	(2.437, 2.634)
Total Effect (TE)	4.5	4.537	(4.457, 4.606)
Proportion Mediated (PM%)	55.56%	55.76%	(53.82%, 57.87%)
TE-sum (verification)	4.5	4.537	(4.457, 4.606)



WHAT IF WE HAVE THE FOLLOWING SETTING?



INVERSE ODDS RATIO WEIGHTING

- Step 1: Logistic regression exposure ~ mediator(s)
 - `glm(female ~ pm + ses, data = dt, family = "binomial")`
- Step 2: Generate IOR for weights
 - `dt$logodds <- predict(e.mod5, type="link", dt)`
 - `dt$inverseodds <- exp(-dt$logodds)`
 - `dt$wt_iow <- ifelse(dt$female==0, 1, dt$inverseodds)`
 - `dt$inverseoddsratio <- 1/(exp(coef(summary(e.mod5))["pm", 1]*dt$pm))`
 - `dt$wt_iorw <- ifelse(dt$female==0, 1, dt$inverseoddsratio)`
- Step 3: Run outcome model for total effects
 - `t.mod5 <- glm(sbp ~ female + ses, data=dt, family="gaussian")`
 - `te5 <- coef(summary(t.mod5))["female", 1] ## total effect`
- Step 4: run weighted outcome models for controlled direct effects
 - `y_iorw.mod5 <- glm(sbp ~ female + ses, data=dt, weights = wt_iorw, family="gaussian")`
 - `de_iorw5 <- coef(summary(y_iorw.mod5))["female", 1] ## controlled direct effect by IORW`
- Step 5: obtained Natural Indirect Effects by contrasting the 2 models
 - `ie_iorw5 <- te5 - de_iorw5 ## natural indirect effect by IORW`

Effect	IORW
Direct Effect	2.598
Indirect Effect	1.939
Total Effect	4.537
Proportion Mediated	0.427





Q&A

Acknowledgements

Chen Chen, Yiqun Ma and the Climate Epidemiology Group
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Sheri Weiser, Arianne Teherani and the P20 team





**PLEASE
COMPLETE THE
SURVEY**

<https://s.zoom.us/m/bPlWfeukn>

THANKS

TBENMARHIA @ UCSD.EDU

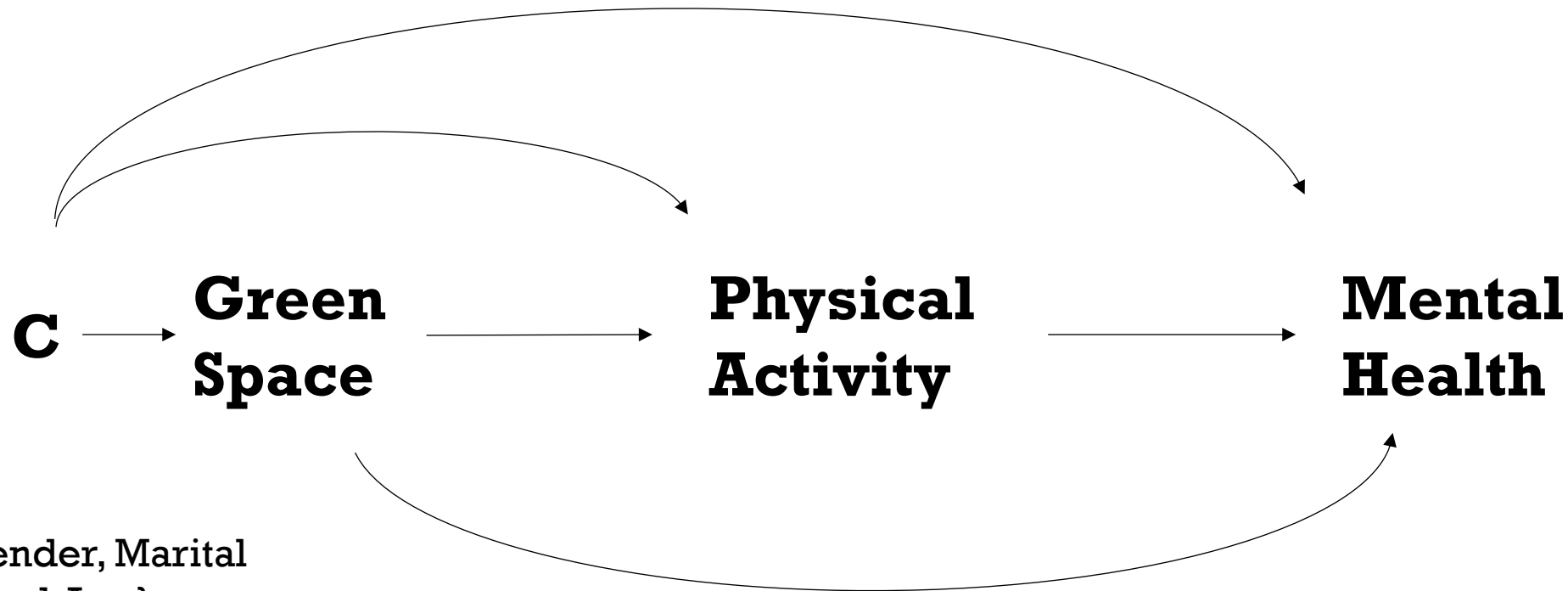


**CASE STUDY:
GREEN SPACE, PHYSICAL ACTIVITY AND
MENTAL HEALTH**



RESEARCH QUESTION

- We want to know the direct effect of Green Space on Mental Health (Depression Scale) and the indirect one through Physical Activity
- Check the assumptions



DATA

- We use some hypothetical cohort:
 - **A is Green space:** Continuous Measure (in %) that represents some Neighborhood green space density
 - **Y is Mental Health:** Continuous Measure, Center for Epidemiologic Studies Depression Scale (CESD) Revised (2004) and reversed.
 - Can be categorized with a cutoff at 16 for Non Depressed (0) vs. Depressed (1);
 - **M is Physical Activity:** Continuous Measure that represents weekly walking (in minutes). Standardized: mean=0
 - **Measured Confounders C** are Gender (Women, Men) and Marital status (Married, not married) and age (continuous measure)
- N= 294 subjects; 111 men and 183 women, mean age = 44 years



“BY HAND” CALCULATION

- **Stage 1:** Estimate the total effect of A (Green Space) on Y (CESD)
- **Stage 2:** Estimate the Outcome model to get CDE & θ_2 (estimate of M on Y)
- **Stage 3:** Estimate the Mediator Model to get β_1 (estimate of A on M)
- **Stage 4:** Calculate the product ($\beta_1 \theta_2$) to get the NIE
- **Stage 5:** Check if $\text{NIE} + \text{CDE} = \text{Total Effect}$



STAGE 1 – TOTAL EFFECT

.regress outcome green sex age marital

outcome	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
green	.1038607	.0341058	3.05	0.003	.0367335	.1709878
sex	-1.840263	1.054916	-1.74	0.082	-3.916554	.2360291
age	.0963298	.0335114	2.87	0.004	.0303725	.1622871
marital	.0698329	.4869196	0.14	0.886	-.8885255	1.028191
_cons	-12.47981	2.432898	-5.13	0.000	-17.26826	-7.691369

- **Total Effect = 0.104 (95%CI: 0.037; 0.171)**
- It means that 1 unit increase in neighborhood green space density is associated with an increase of 0.104 units in well being (positive mental health: reversed CESD)



STAGE 2 – OUTCOME MODEL

CONTROLLED DIRECT EFFECT

.regress outcome green PA sex age marital

outcome	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
green	.0846156	.0333145	2.54	0.012	.0190448	.1501863
PA	2.304501	.5149851	4.47	0.000	1.290889	3.318112
sex	-1.489616	1.024819	-1.45	0.147	-3.506701	.5274685
age	.1375545	.033742	4.08	0.000	.0711424	.2039667
marital	-.0829435	.4728772	-0.18	0.861	-1.013677	.8477899
_cons	-14.12104	2.384938	-5.92	0.000	-18.81516	-9.426917

θ_1

θ_2

- **CDE = 0.085 (95%CI: 0.019; 0.150)**
- The effect of green space on mental health, after intervening to fix the level of physical activity to a certain value (here mean=0) is thus 0.085.
- We save θ_2 (estimate of M on Y) = 2.305 (95%CI: 1.291; 3.318)



STAGE 3: MEDIATOR MODEL

. regress PA green sex age marital

PA	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
green	.0083511	.0037735	2.21	0.028	.0009241	.0157781
sex	-.1521572	.116716	-1.30	0.193	-.3818783	.077564
age	-.0178888	.0037077	-4.82	0.000	-.0251863	-.0105913
marital	.0662948	.0538728	1.23	0.219	-.0397381	.1723277
_cons	.7121812	.2691761	2.65	0.009	.182387	1.241975

- β_1 (estimate of A on M) = 0.008 (95%CI: 0.001; 0.016)



STAGE 4: ESTIMATE THE NATURAL INDIRECT EFFECT

- Calculate the product ($\beta_1 \times \theta_2$)
 - $NIE = 0.008 \times 2.305$
 - **NIE = 0.019**
- *Which represents the change in mental health when the level of green space is held constant ($\bar{A}=a$), and *physical activity* changes to what it would have taken for one unit increase in green space ($\bar{A}=a^*$).*
- But, we **do not** have Confidence Intervals for the NIE..



STAGE 5: CHECK

$NIE + CDE = \text{TOTAL EFFECT ?}$

- Total Effect = **0.104** (from Stage 1)
- $CDE = 0.085$
- $NIE = 0.019$

$$CDE + NIE = 0.085 + 0.019 = \mathbf{0.104}$$



WRITING A PROGRAM IN STATA TO GET CONFIDENCE INTERVALS

- Let's write a program that we will call: mediation_cont_cont
- We will get Confidence Intervals using Bootstrapping
- We will store regressions estimates using the “Scalar command”

```
/******
```

```
// Manual Mediation Program with bootstrapping
```

```
// CONTINUOUS EXPOSURE, CONTINUOUS MEDIATOR, CONTINUOUS OUTCOME
```

```
// Variables (parameters) sent to program:
```

```
// 1 = EXPOSURE --> GREEN SPACE
```

```
// 2 = MEDIATOR --> PHYSICAL ACTIVITY
```

```
// 3 = OUTCOME --> CESD
```

```
// 4 - 6 = covariates C1-C3 --> GENDER, AGE, MARITAL STATUS
```

```
*****
```



OUTCOME MODEL (1/4)

```

/*****
program mediation_cont_cont, rclass
// outcome model                                E[Y | a,m,c]
    regress `3' `1' `2' `4' `5' `6'
    scalar AonY = _b[`1']
    scalar MonY = _b[`2']

    return scalar cde = AonY                                // Controlled Direct Effect
                                                                E[CDE(m*) | C]
*****/

```



MEDIATION MODEL (2/4)

```

/*****/
// mediation model                                 $E[M | a, c]$ 
    regress `2' `1' `4' `5' `6'
    scalar intercept = _b[_cons]
    scalar b_c1 = _b[`4']
    scalar b_c2 = _b[`5']
    scalar b_c3 = _b[`6']
    scalar AonM = _b[`1']

    return scalar nie = AonM*MonY                                // Natural Indirect
                                                                    Effect
    return scalar te = (AonM*MonY)+AonY                            // Total Effect
/*****/

```



CONFOUNDERS (3/4)

```

/*****/
    summarize `4'
    scalar m_c1 = r(mean)
    summarize `5'
    scalar m_c2 = r(mean)
    summarize `6'
    scalar m_c3 = r(mean)

end
/*****/
```



GET THE CONFIDENCE INTERVALS (4/4)

/*****/

bootstrap CDE = r(cde), reps(1000) : mediation_cont_cont green PA outcome sex age marital

bootstrap NIE = r(nie), reps(1000) : mediation_cont_cont green PA outcome sex age marital

bootstrap TE = r(te), reps(1000) : mediation_cont_cont green PA outcome sex age marital

/*****/



RESULTS

Bootstrap results

Number of obs = 294
Replications = 1,000

	Observed Coef.	Bootstrap Std. Err.	z	P> z	Normal-based [95% Conf. Interval]	
CDE	.0846156	.0261376	3.24	0.001	.0333869	.1358442

	Observed Coef.	Bootstrap Std. Err.	z	P> z	Normal-based [95% Conf. Interval]	
NIE	.0192451	.0105089	1.83	0.067	-.001352	.0398422

	Observed Coef.	Bootstrap Std. Err.	z	P> z	Normal-based [95% Conf. Interval]	
TE	.1038607	.0311123	3.34	0.001	.0428817	.1648396

